

NAME(S):

Real-World Recursion

Begin the following as paired-programming in class, but be sure to individually finish any problems you don't get to (hand in written answers of **0.25 engagement points**, check website for due date).

For each scenario below, discuss with your lab partner and write:

- Base case: When does the recursion stop?
- Recursive step: How do you break the problem into a smaller version of itself?

Example format:

Scenario: "How many socks are in a pile?"

- Base case: If the pile is empty, return 0.
- Recursive step: Otherwise, return 1 + the number of socks in a pile that has 1 less sock.

Scenarios

1. How many slices are in a loaf of bread?

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2. How many steps in a staircase?

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3. How many people are in a line at the grocery store?

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4. How many pages are in a book?

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5. How many cups are in a stack?

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